Ashlesha Singh

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EDUCATION

Symbiosis Institute of Technology, Nagpur 2024 – 2028
Bachelor of Technology in Computer Science and Engineering CGPA: 8.45

- Specialization: AIML | Batch: 2024–28

PERSONAL PROJECTS

Vidarbha Urbanization Impact Analysis July 2025 – Nov 2025

- Architected a data pipeline for spatio-temporal analysis to quantify urbanization impacts across 11 districts over 20 years
- Implemented a Random Forest model mapping Land Use and Land Cover (LULC), identifying 7,100+ sq km of forest loss and 800+ sq km of urban growth
- Analyzed satellite datasets (Landsat, MODIS) using 500 annual spatial samples to track environmental degradation
- Quantified the Urban Heat Island effect and a $>$100% increase in PM2.5 pollution via statistical testing
- **Technologies Used:** Python, JavaScript, Google Earth Engine, Random Forest, Pandas, NumPy, SciPy, Rasterio, Matplotlib, Seaborn

Strategic Afforestation Dashboard Jan 2026 – Present

- Developed a Decision Support System (DSS) to eliminate random planting, identifying optimal, data-driven afforestation zones for urban heat mitigation in Nagpur
- Formulated a Multi-Criteria Decision Analysis (MCDA) engine and spatial mask (via Google Open Buildings) to prioritize viable ground and rooftop areas based on heat, greenery need, and population metrics
- Constructed a frontend dashboard with interactive weight sliders and real-time mapping layers to enable smart city planning
- **Technologies Used:** Google Earth Engine (GEE), JavaScript, Multi-Criteria Decision Analysis (MCDA), Google Open Buildings Dataset

Smart Traffic Light System Sep 2025

- Engineered an adaptive traffic control system leveraging YOLOv8 and OpenCV to dynamically adjust green light durations based on real-time vehicle density
- Processed video feeds frame-by-frame to accurately detect, classify, and count vehicles, categorizing traffic flow into dynamic thresholds (Low, Medium, High)
- Replaced static signal timers with an automated algorithm, optimizing intersection throughput by scaling light durations from 15 to 60 seconds to minimize congestion
- **Technology Used:** Python, YOLOv8, OpenCV, Google Colab

Digital Password Authentication System

- Designed a 4-bit binary authentication circuit employing combinational logic (XNOR and AND gates) to verify user credentials against a preset hardware key
- Fabricated a functional PCB prototype utilizing 74LS266 (Quad XNOR) and 74LS08 (Quad AND) ICs for bitwise comparison and final authorization logic
- Validated hardware security and integrity through rigorous testing using SPST switches and LED indicators to confirm logic state accuracy
- **Technologies Used:** Combinational Logic Design, PCB Fabrication, Digital ICs (74LS266, 74LS08), Hardware Testing

TECHNICAL SKILLS AND INTERESTS

Languages: Python, JavaScript, C, HTML, CSS, LaTeX
Tools: VS Code, Google Colab, Google Earth Engine (GEE), GitHub
Libraries & Frameworks: YOLOv8, Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, Seaborn
Areas of Interest: AI & ML, Geospatial Data Science, Web Design & Development, Digital Electronics

CERTIFICATIONS & ACHIEVEMENTS

Red Hat Certified Specialist in Python Programming (AD141) March 2025

Data Structures and Algorithms using Python Oct 2025
Infosys Springboard

Data Science Summer School (DS3) Participant Aug 2025
Hertie School Data Science Lab

Game Development Workshop Dec 2025
IIT Bombay Techfest

Introduction to Retrieval-Augmented Generation (RAG)
IBM SkillsBuild

Ethical Considerations for Generative AI
IBM SkillsBuild